Exam Date & Time: 04-Jan-2023 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - JANUARY 2023
SUBJECT: BIO 1071 / BIO-1071 - BIOLOGY FOR ENGINEERS

BIOLOGY FOR ENGINEERS [BIO 1071]

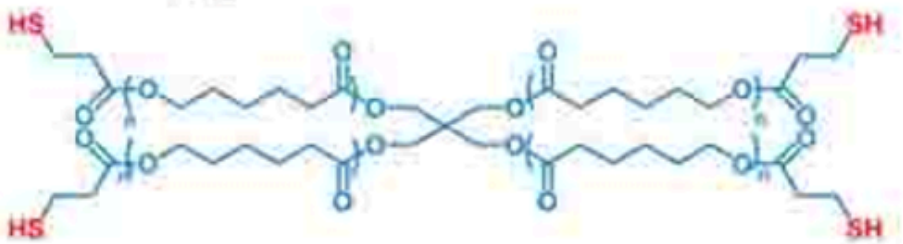
Marks: 50 **Duration: 180 mins.**

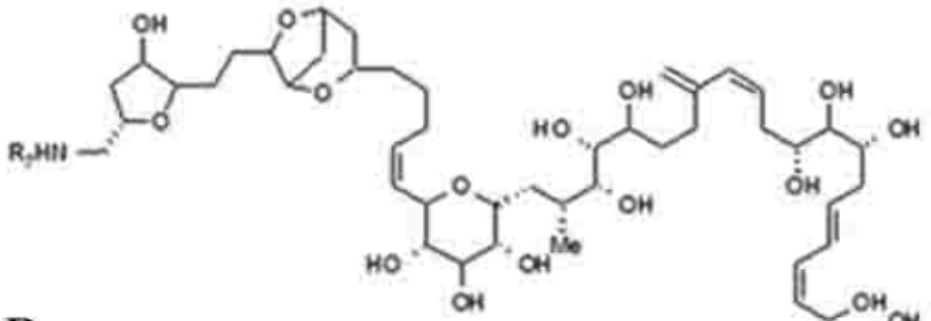
A

Answer all the questions.

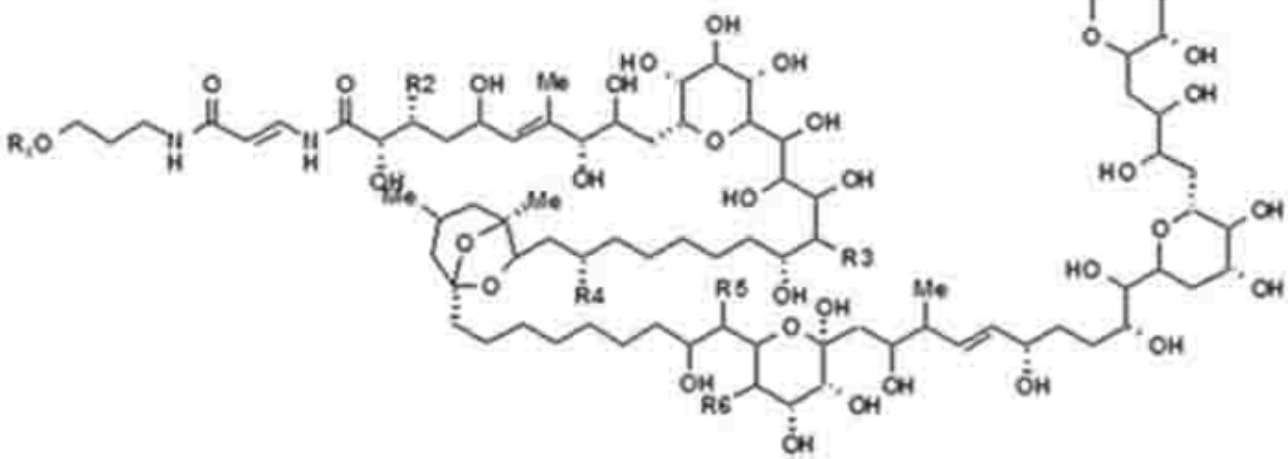
1A) After graduation from MIT, you are working in an industry. You are considering the following compounds A and B for hydrophilic and lipophilic properties and required to submit a report to your boss.

A





B



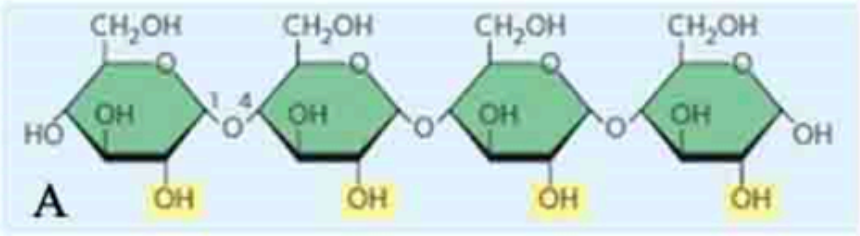
(3)

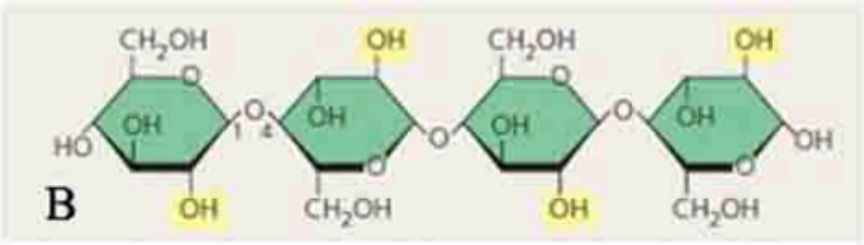
Prepare your report in the following format

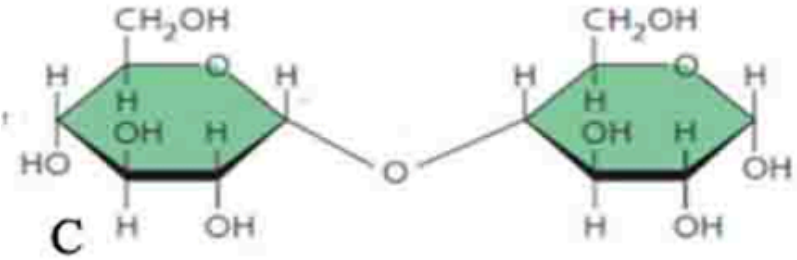
Compound	Hydrophilic?	Lipophilic?	Justification
A			
B			

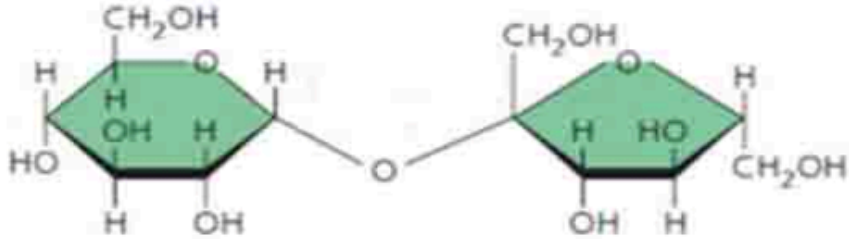
1B) A The International Space Station menu provides ~50% of calories as carbohydrate, 17% as protein, and 31% as fat. According to you, which of the following compound can be included in (4)

carbohydrates menu? Why the others can't be included? Justify.







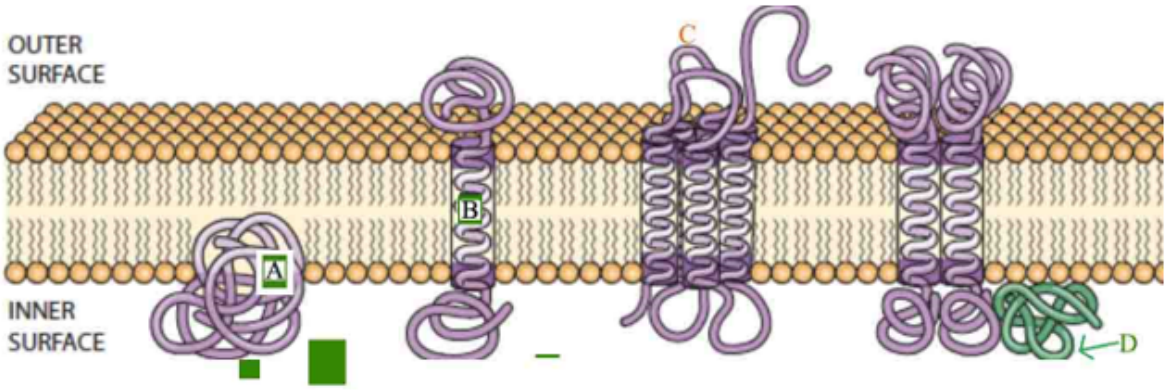


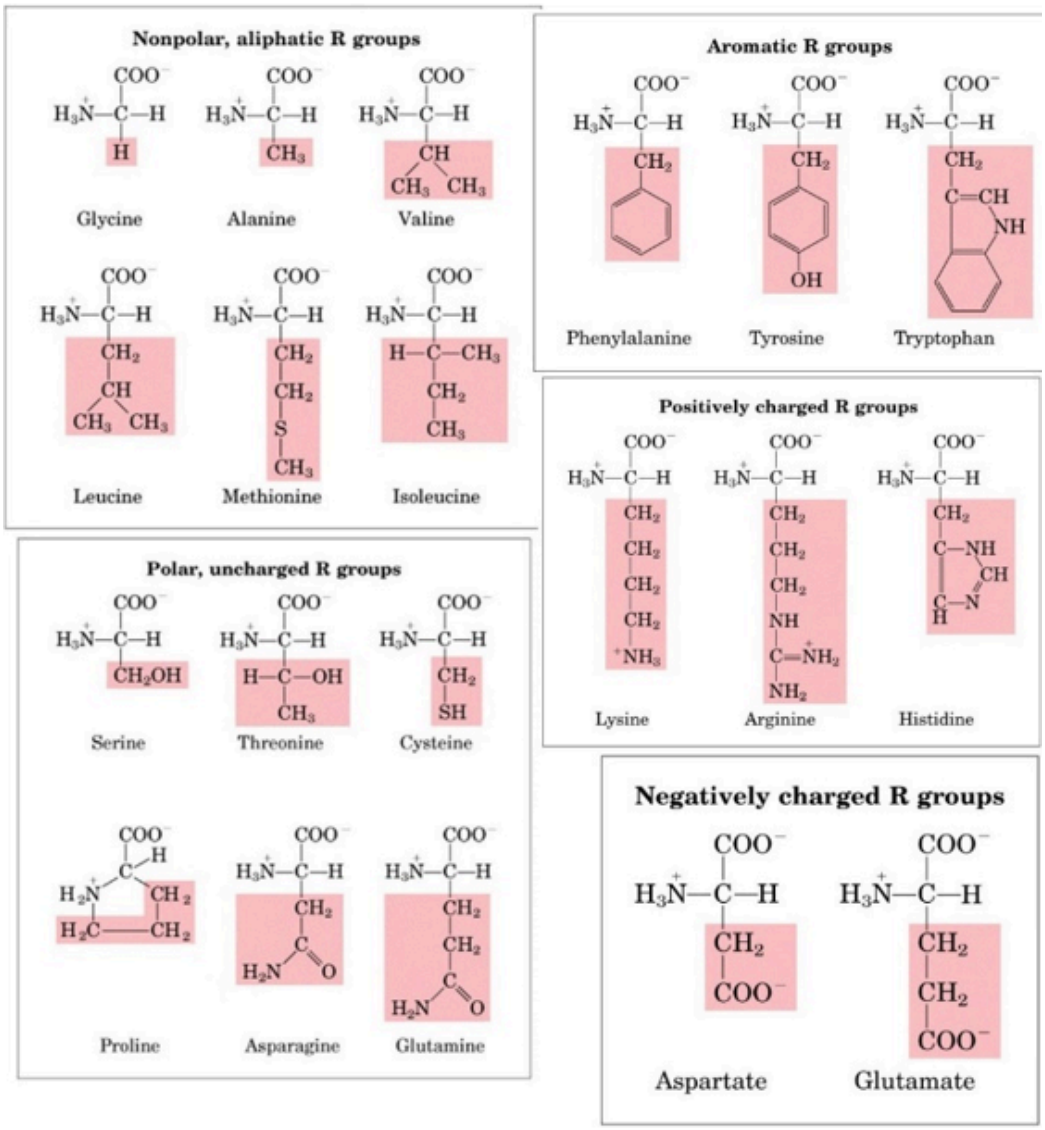
D

You can copy the table and fill the respective columns for your response

Source	The type of glycosidic linkage	Can it be included in carbohydrate menu? Justification
A		
B		
C		
D		

1C) Given illustration below explains the existence of protein in a cell membrane. Mr Keerthan has discovered that the following amino acids can be substituted to the regions labelled as A, B and C without disturbing the structure and function, but improves the stability. (3)





(A) According to you which amino acid is most suitable for each position (among the given options - Tryptophan, Threonine, Glutamine) and why?
You can submit your answer by filling the table

Label	Your amino acid choice with justification
A	
B	
C	

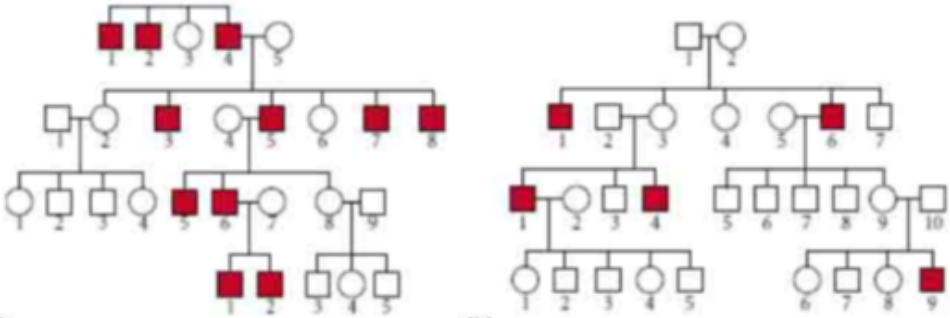
(B) What is the type of protein structure is visible at point D?
(C) Mr Keerthan replaced all the lysine with glycine before submitting to his professor. Will that protein be still be functional? Justify.
(D) Mr Keerthan exposed the protein with some chemical and got denatured ie lost its structure. He informed his professor and he says that it can't be reversed once lost the protein structure. Based on your understanding give a proper justification.
(1½+½+½+½ = 3 marks)

2A) Predict the outcome of the cross in the following:
Tall Round (TtRr) X dwarf wrinkled (ttrr)
Calculate phenotypic and genotypic ratio (3)

2B) How Morgan was able to prove that the factor for eye colour is located on X chromosome. Logically explain. (3)

2C) (4)

Identify the most probable mode of inheritance in (a) and (b). Justify.



(a) (b)

What is the probability of a male being affected for an affected father in this mode of inheritance (a) and (b)?

3A) A RNA polymerase is involved in transcribing a DNA segment (given below). If the polymerase transcribes this sequence from left to right, what will the sequence of RNA be? What will the RNA sequence be if the polymerase moves right to left? For the both the cases what would be the template strand. (3)

5′-GTAACGGATG-3′
3′-CATTGCCTAC-5′

3B) A group of students wanted to repeat the experiments conducted by Meselson-Stahl. The bacteria was grown initially in the heavy nitrogen media and then shifted to lighter nitrogen media. The students conducted density gradient centrifugation to deduce the mode of replication of this newly found bacteria. What mode of replication does this bacteria follow ?
How will you logically rule out other modes of replication here?
L = Light strand ^{14}N and H= Heavy strand ^{15}N

(3)

(1+2 = 3 marks)

3C) DNA sequence shown below is of a gene (only one strand sequence been given) that encodes a short peptide. Peptide sequence is also given below. Based on the information given answer the following question.
3'-TCGA GTAC ACG CTC AGG ACT GCG ACT GATCC-5'
NH_3^+ Met-Cys-Glu-Arg-Coo^- (4)

i) Give the mature mRNA sequence of the gene with proper labels - newly formed and mature.
ii) Justify why only 4 amino acid are synthesized from 31 nucleotides?
(2+1 = 3 marks)

4A) (i) Hummingbirds feed on nectar from flowers. The flowers are pollinated by hummingbirds as they move from flower to flower. What type of interaction is seen in this?
(ii) Biofilms can tolerate antimicrobials and is drug resistant. Can the biofilm be mutual benefit or threat to community? Justify? (3)
(1+2 = 3 marks)

4B) A small change in a molecule has led to the diverse functions. For this we need to examine molecules such as haemoglobin, myoglobin, hemocyanin, leghaemoglobin and chlorophyll. All of them are modified porphyrins. (3)
(i) What is the central metal ion coordinated to Porphyrin ring in case of Chlorophyll?
(ii) In case of hemocyanin crabs, octopus, snails, lobsters have blue blood, why?

(iii) What is the structural difference between Haemoglobin and Myoglobin?
(1+1+1 = 3 marks)

4C) Several attempts have been made from time to time to explain the origin of life on earth. As a result, there are several theories which offer their own explanation on the possible mechanism of origin of life. How can the justification given for the following theories, for origin of life?
(i) Theory of Special Creation
(ii) Theory of Spontaneous Generation (4)
(iii) Theory of Biogenesis
(iv) Theory of Biochemical Evolution
(1 mark × 4 = 4 marks)

5A) Give one example of living organisms which Bioinspired to create the following technology.
(i) Water Filters
(ii) Shock absorber (3)
(iii) Bionic Car

5B) What does the following Symbols represent?
(i) +/0
(ii) +/- (3)
(iii) +/+

5C) (i) Vaccines can induce immunity. How is it possible?
(ii) Inactivated microorganism are used in the form of vaccines. Justify.
(iii) Is the booster shot required for all types of diseases? (4)
(iv) How adjuvants have the capacity to increase effectiveness of vaccines?

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